

PHY 555: Solid-state Physics I

Homework #3

Due: 10/06/2025

Homework is due by the end of the due date specified above. **Late homework will be subject to 3 points off per day past the deadline, please contact me if you anticipate an issue making the deadline.** It should be turned in via blackboard. For the conceptual and analytical parts, turn in a scan or picture of your answers (please ensure that they are legible) or an electronic copy if done with, e.g., $\text{\LaTeX}$. For the computational part, turn in your source code and a short description of your results (including plots). The description can be separate (e.g., in $\text{\LaTeX}$ or word), or combined (e.g., in a jupyter notebook). Let me know if you are not sure about the format.

Conceptual

1. **5 points** What are the approximations made to go from the expression for the fully general tight-binding matrix elements and secular equation:

$$\begin{aligned} M_{ij\mathbf{k}} &= \langle \Phi_{i\mathbf{k}} | H | \Phi_{j\mathbf{k}} \rangle \\ S_{ij\mathbf{k}} &= \langle \Phi_{i\mathbf{k}} | \Phi_{j\mathbf{k}} \rangle \\ \det | M_{ij\mathbf{k}} - E S_{ij\mathbf{k}} | &= 0 \end{aligned}$$

to the semi-empirical expressions:

$$\begin{aligned} M_{ij\mathbf{k}} &= E_i \delta_{ij} + \sum_{\mathbf{t}_I} e^{i\mathbf{k} \cdot \mathbf{t}_I} \int \phi_i^*(\mathbf{r}) V_a(\mathbf{r} - \mathbf{t}_I) \phi_j(\mathbf{r} - \mathbf{t}_I) d^3r \\ S_{ij\mathbf{k}} &= \delta_{ij} \\ \det | M_{ij\mathbf{k}} - E \delta_{ij} | &= 0 \end{aligned}$$

where $\Phi_{i\mathbf{k}}$'s are the Bloch sums, ϕ_i 's are the atomic-like orbitals, $\mathbf{t}_I$ runs over nearest neighbor lattice sites, and $V_a(\mathbf{r} - \mathbf{t}_I)$ is the atomic-like potential centered on site $\mathbf{t}_I$.

Analytical

2. **20 points** Consider the empirical tight-binding treatment of s and p electrons on an FCC lattice. In this case, each lattice site has twelve nearest neighbors $\mathbf{t}_I$: $(a/2)(0, \pm 1, \pm 1)$, $(a/2)(\pm 1, 0, \pm 1)$, $(a/2)(\pm 1, \pm 1, 0)$. The relevant interaction integrals are:

$$\begin{aligned} \int \phi_s^*(\mathbf{r}) V_a(\mathbf{r} - \mathbf{t}_I) \phi_s(\mathbf{r} - \mathbf{t}_I) d\mathbf{r} &= V_{ss\sigma} \\ \int \phi_s^*(\mathbf{r}) V_a(\mathbf{r} - \mathbf{t}_I) \phi_{p_x}(\mathbf{r} - \mathbf{t}_I) d\mathbf{r} &= l_x V_{sp\sigma} \\ \int \phi_{p_x}^*(\mathbf{r}) V_a(\mathbf{r} - \mathbf{t}_I) \phi_{p_x}(\mathbf{r} - \mathbf{t}_I) d\mathbf{r} &= l_x^2 V_{pp\sigma} + (1 - l_x^2) V_{pp\pi} \\ \int \phi_{p_x}^*(\mathbf{r}) V_a(\mathbf{r} - \mathbf{t}_I) \phi_{p_y}(\mathbf{r} - \mathbf{t}_I) d\mathbf{r} &= l_x l_y (V_{pp\sigma} - V_{pp\pi}) \\ \int \phi_{p_x}^*(\mathbf{r}) V_a(\mathbf{r} - \mathbf{t}_I) \phi_{p_z}(\mathbf{r} - \mathbf{t}_I) d\mathbf{r} &= l_x l_z (V_{pp\sigma} - V_{pp\pi}) \end{aligned}$$

where ϕ_i 's are the atomic-like orbitals; $V_{ss\sigma}$, $V_{sp\sigma}$, $V_{pp\sigma}$, and $V_{pp\pi}$ are the adjustable parameters; and $\mathbf{l} = (l_x, l_y, l_z)$ is the unit vector in the direction of $\mathbf{t}_I$.

- (a) Assume that the s orbitals do not interact with the p orbitals. Show that the dispersion of the s -orbital derived band is given by

$$E(\mathbf{k}) = E_s + 4V_{ss\sigma} \left[\cos\left(\frac{k_x a}{2}\right) \cos\left(\frac{k_y a}{2}\right) + \cos\left(\frac{k_y a}{2}\right) \cos\left(\frac{k_z a}{2}\right) + \cos\left(\frac{k_z a}{2}\right) \cos\left(\frac{k_x a}{2}\right) \right]$$

where E_s are the onsite energies.

- (b) For the p orbital manifold, show that the matrix elements are given by:

$$\begin{aligned} M_{p_x p_x \mathbf{k}} &= E_p + 2 \cos\left(\frac{k_x a}{2}\right) \left[\cos\left(\frac{k_y a}{2}\right) + \cos\left(\frac{k_z a}{2}\right) \right] (V_{pp\sigma} + V_{pp\pi}) \\ &\quad + 4 \cos\left(\frac{k_y a}{2}\right) \cos\left(\frac{k_z a}{2}\right) V_{pp\pi} \\ M_{p_x p_y \mathbf{k}} &= -2 \sin\left(\frac{k_x a}{2}\right) \sin\left(\frac{k_y a}{2}\right) (V_{pp\sigma} - V_{pp\pi}) \end{aligned}$$

Computational

3. **25 points** Use the results of problem 2 to calculate the dispersion of the s and p bands in an FCC crystal along the path $L \rightarrow \Gamma \rightarrow X \rightarrow K \rightarrow \Gamma$ (see Fig. 1 and Table 1 for high-symmetry $\mathbf{k}$ points/paths). Use the following parameters (all in Ha): $V_{ss\sigma} = -0.5$, $V_{pp\sigma} = 0.5$, $V_{pp\pi} = -0.05$, $E_s = 9$, $E_p = 0$. Use $a = 10.67$ Bohr for the lattice constant.

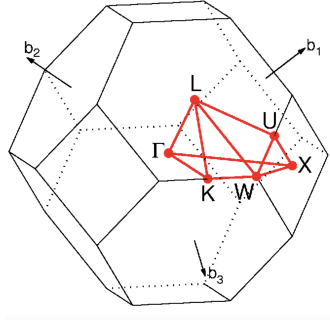


Figure 1: Brillouin zone and high-symmetry points/lines for the face-centered cubic Bravais lattice. (b_1 , b_2 , and b_3 correspond to the reciprocal lattice vectors of FCC, written below as $\mathbf{g}_1$, $\mathbf{g}_2$, and $\mathbf{g}_3$.)

Table 1: High-symmetry $\mathbf{k}$ points of the face-centered cubic lattice.

	$\times \mathbf{g}_1$	$\times \mathbf{g}_2$	$\times \mathbf{g}_3$
Γ	0	0	0
K	3/8	3/8	3/4
L	1/2	1/2	1/2
U	5/8	1/4	5/8
W	1/2	1/4	3/4
X	1/2	0	1/2